

2.1.

$$6.50, \quad \langle x, y \rangle = [x]_B^* [y]_B$$

$$[id]_B^K = ([\vec{k}_1]_B, [\vec{k}_2]_B)$$

$$\left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 2 & 6 & 2 & 0 \\ -2 & -7 & 0 & -1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & -1 & 2 & -1 \end{array} \right)$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b_1 \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$-b_1 = 2$$

$$a_1 = 1 - 3b_1$$

$$-b_2 = -1$$

$$a_1 = -3b_2$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = a_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b_2 \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$b_1 = -2$$

$$a_1 = 1 + 6 = 7$$

$$b_2 = 1$$

$$a_1 = -3$$

$$[id]_B^K = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$$

$$[x]_B = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix}$$

$$[x]_B^* = (7x_1 - 3x_2 \quad -2x_1 + x_2)$$

$$[y]_B = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix}$$

$$\langle x, y \rangle = (7x_1 - 3x_2 \quad -2x_1 + x_2) \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix} =$$

$$= (7x_1 - 3x_2)(7y_1 - 3y_2) + (-2x_1 + x_2)(-2y_1 + y_2) =$$

$$= 49x_1y_1 - 21x_1y_2 - 21x_2y_1 + 9x_2y_2 + 4x_1y_1 - 2x_1y_2 - 2x_2y_1 + x_2y_2 =$$

$$= 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 10x_2y_2$$